

Implement the ID3, C4.5, and CART decision tree algorithms from scratch (without using any machine learning libraries) for a suitable classification dataset. Develop the complete training procedure by implementing the required attribute selection measures, construct the decision trees, and use the generated models to classify test instances. Compare the three algorithms in terms of classification accuracy, precision, recall, F1-score, tree depth, number of leaf nodes, and training time, and summarize your observations on their performance and the resulting tree structures.